COM111 Tutorial 4a

1. The voltage across a capacitor 4 μF is shown in Fig. 1. Find the current waveform.

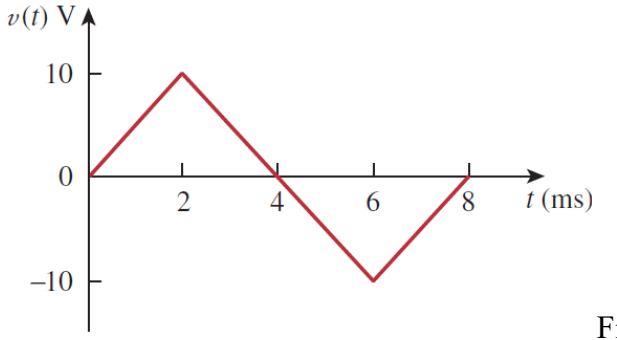


Fig. 1

2. At t = 0, the voltage across a 25-mF capacitor is 10 V. Calculate the voltage across the capacitor for t > 0 when current 5t mA flows through it.

3. A 4-mF capacitor has the terminal voltage

$$v = \begin{cases} 50\ \text{V}, & t \le 0 \\ Ae^{-100t} + Be^{-600t}\ \text{V}, & t \ge 0 \end{cases}$$

If the capacitor has an initial current of 2 A, find:

(a) the constants A and B,

(b) the energy stored in the capacitor at t = 0,

(c) the capacitor current for t > 0.

4. The current through a 0.5-F capacitor is 6 (1 - e^{-t}) A. Determine the voltage and power at t = 2s. Assume v(0) = 0.

5. Find the voltage across the capacitors in the circuit of Fig. 2 under dc conditions.

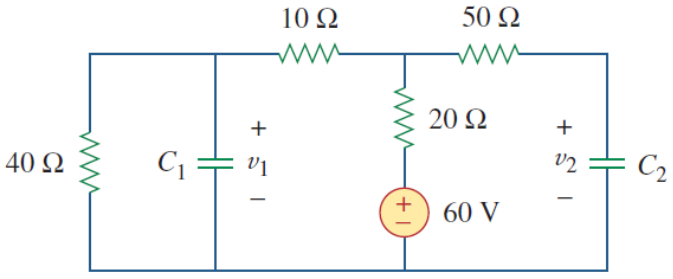


Fig. 2

6. Determine the equivalent capacitance for each of the circuits of Fig. 3.:

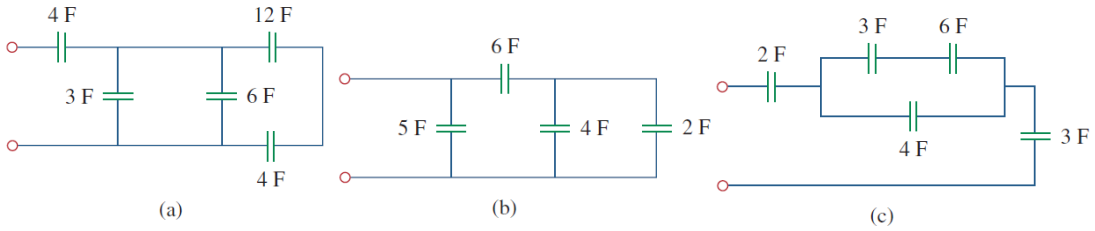


Fig. 3

7. Assuming that the capacitors are initially uncharged, find vo(t) in the circuit of Fig. 4.

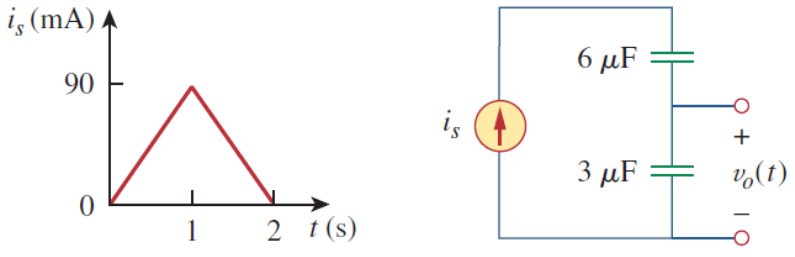


Fig. 4

8. The current through a 40-mH inductor is

$$i(t) = \begin{cases} 0, & t < 0 \\ te^{-2t}\ \text{A}, & t > 0 \end{cases}$$

Find the voltage v(t).

9. A 100-mH inductor is connected in parallel with a 2-kΩ resistor. The current through the inductor is $50e^{-400t}$ mA.

(a) Find the voltage v_L across the inductor.

(b) Find the voltage v_R across the resistor.

(c) Does $v_R + v_L = 0$?

(d) Calculate the energy in the inductor at t = 0.

10. Find *L*eq at the terminals of the circuit in Fig. 5.

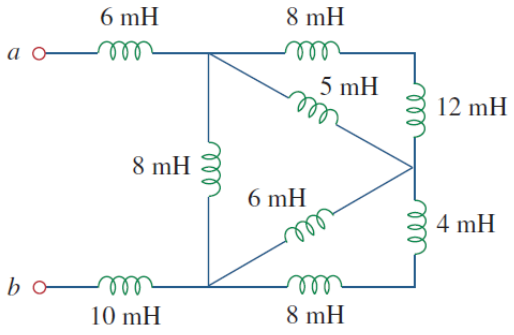


Fig. 5

11. Consider the circuit in Fig. 6. Find:

(a) The L_{eq}, i1(t), and $i_2(t)$ if $i_s = 3e^{-t}$ mA.

(b) The output voltage $v_o(t)$.

(c) The energy stored in the 20-mH inductor at t = 1 s.

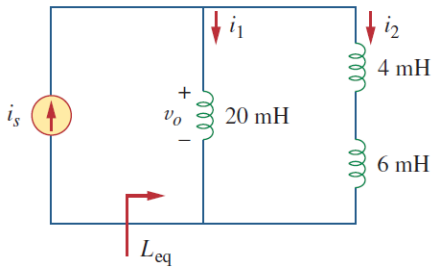


Fig. 6

12. Find the time constant for the *RC* circuit in Fig. 7.

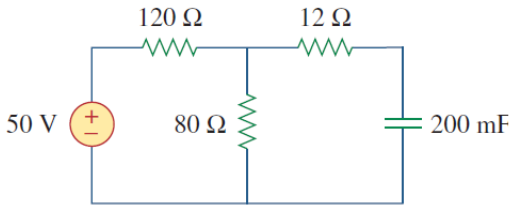


Fig. 7

13. The switch in Fig. 8 has been in position A for a long time. Assume the switch moves instantaneously from A to B at t = 0. Find v for t > 0.

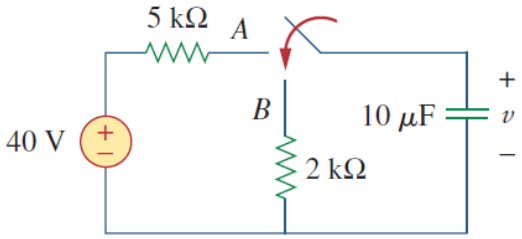


Fig. 8

14. The switch in Fig. 9 has been closed for a long time, and it opens at t = 0. Find v(t) for t > 0.

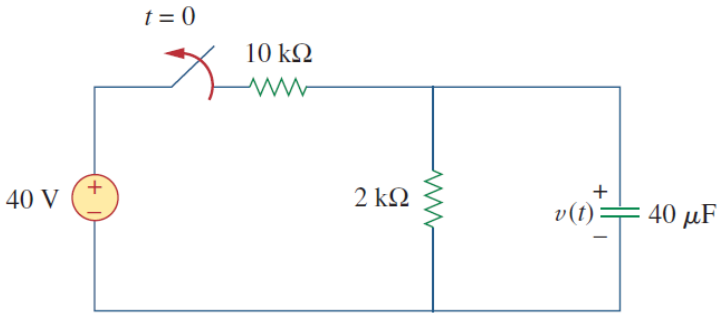


Fig. 9

15. For the circuit in Fig. 10, find *i*o for *t* > 0.

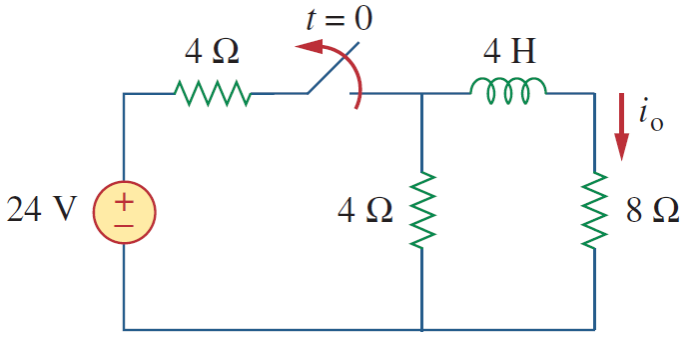


Fig. 10

16. Find the time constant for each of the circuits in Fig. 11.

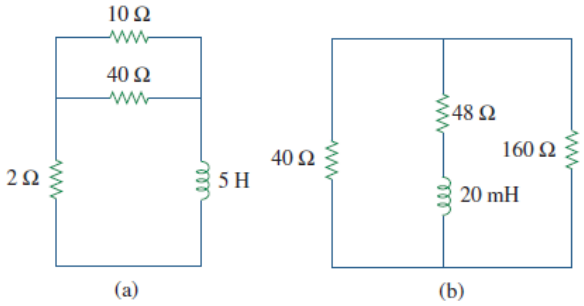


Fig. 11

17. Calculate the capacitor voltage for t < 0 and t > =for each of the circuits in Fig. 12.

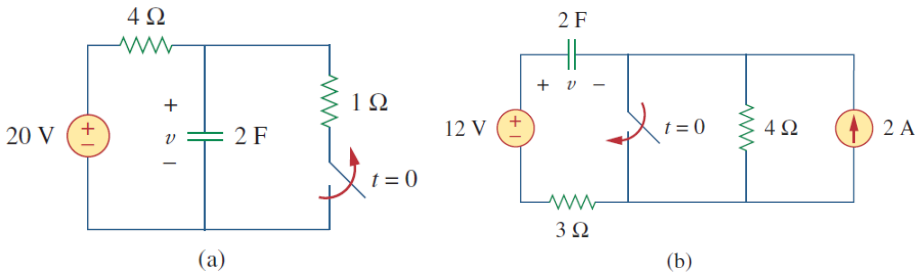


Fig. 12

18. Consider the circuit in Fig. 13. Find i(t) for t < 0 and t > 0.

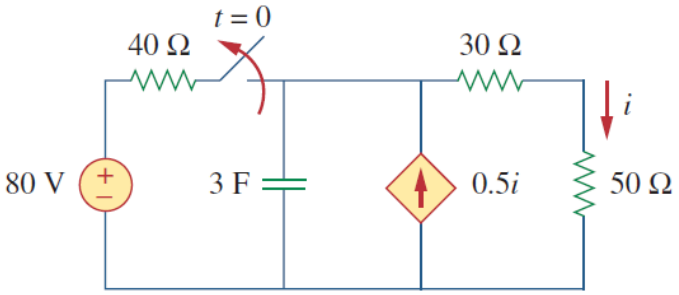


Fig 13

19. The switch in Fig. 14 has been in position a for a long time. At t = 0, it moves to position b. Calculate i(t) for all t > 0.

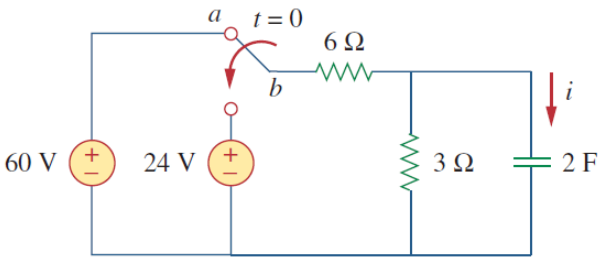


Fig. 14

20. Determine the inductor current i(t) for both t < 0 and t > 0 for each of the circuits in Fig. 15.

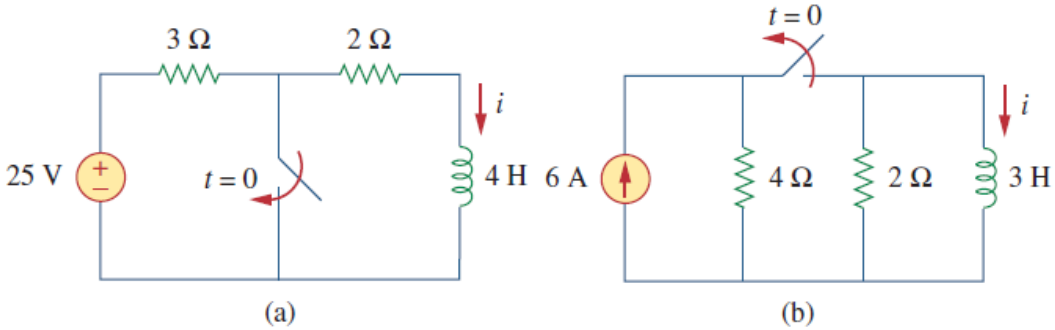


Fig. 15

21. Obtain the inductor current for both t < 0 and t > 0 in each of the circuits in Fig. 16.

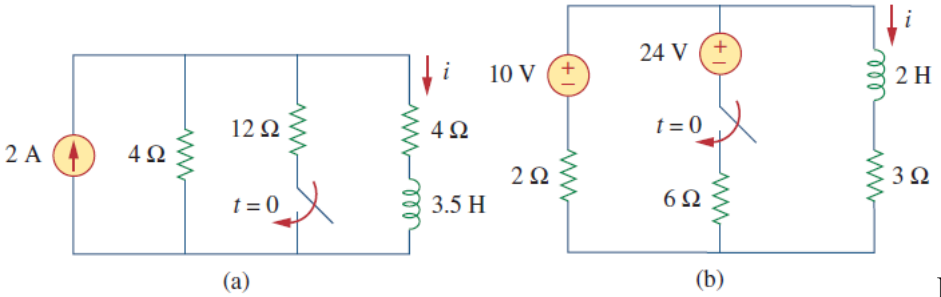


Fig. 16

22. For the network shown in Fig. 17, find v(t) for t > 0.

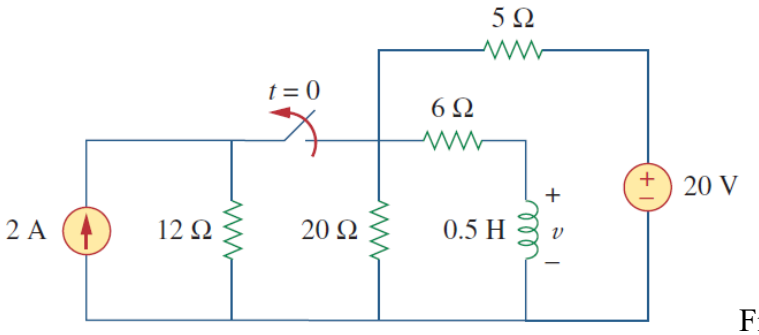


Fig. 17